Study Notes: Entry, Exit, and the Determinants of Market Structure

Dunne, Klimek, Roberts, and Xu (RAND J. Econ., 2013)

Contents

1 What the Paper Does (One Coherent Story)

This paper estimates a *dynamic structural* model of market structure where

- **Incumbents** decide whether to continue or exit, facing **fixed operating costs**.
- **Potential entrants** decide whether to enter, facing **sunk entry costs**.
- Short-run profits depend on market state (especially the number of incumbents), capturing the **toughness of competition**.

The key idea is to treat three objects as the model primitives:

1. the reduced-form **profit function** $\pi(s)$ (competition effect is primarily how π falls with n),
2. the **fixed cost distribution** G_λ governing exit,
3. the **entry cost distribution** G_κ governing entry.

The application uses two U.S. health service industries:

- **Dentists**, where some markets are eligible for entry subsidies via **Health Professional Shortage Areas (HPSA)** designation.
- **Chiropractors**, used as a comparison industry with different observed turnover and profit patterns.

The empirical question is: *How do competition, fixed costs, and entry costs jointly determine long-run firm values, turnover (entry/exit), and market structure?*

2 Data and Institutional Setting

2.1 Market Definition and Timing

Markets are small geographic units (incorporated census places), observed at five-year intervals:

$$t \in \{1982, 1987, 1992, 1997, 2002\}.$$

For each market m and time t, we observe:

- n_{mt}: number of incumbent establishments (active practices) at the beginning of the period,
- e_{mt}: number of entrants over the previous five years,
- x_{mt}: number of exits over the previous five years,
- a set of exogenous shifters z_{mt} (demand and cost variables),
- an average profit measure (net income per owner-practitioner, constructed from Census sources).

The number of firms evolves as

$$n' = n + e - x.$$

2.2 Profit Measurement and the Five-Year Step

Profit is constructed as *net income per owner-practitioner* (with corrections for legal form and owner compensation). Because entry and exit are measured over five-year intervals, the paper converts annual profits into a discounted five-year flow and then uses a five-year discount factor in the dynamic problem:

- annual discount factor: $\delta_{\text{annual}} = 0.95$,
- effective five-year discount factor in the dynamic model:

$$\delta \equiv (0.95)^5 \approx 0.773.$$

2.3 Interpreting Table 1: Demand, Structure, and Turnover

Table 1 reports market averages by population quartiles for:

- dentists in non-HPSA markets,
- dentists in HPSA markets,
- chiropractors.

Main facts from Table 1.

- **Dentist non-HPSA markets:** n rises strongly with population (from 3.86 in Q1 to 11.90 in Q4). Entry and exit proportions are stable around ≈ 0.20.
- **Dentist HPSA markets:** conditional on population, there are typically **fewer firms** (e.g., Q4: 8.55 vs 11.90). Turnover is often higher (e.g., Q3 entry proportion 0.285 vs 0.206 in non-HPSA).
- **Chiropractors:** far fewer firms (Q4 only 3.84) and dramatically higher entry proportions (around 0.41–0.52) with exit rates around 0.23–0.25.

Economic interpretation. These patterns suggest that dentists and chiropractors differ in:

- market profitability (demand strength and/or prices),
- fixed costs (exit incentives),
- entry costs (entry incentives),
- and potentially the competitive effect of additional firms on profits.

The model is designed precisely to separate these components.

3 Baseline Dynamic Entry–Exit Model

The model follows the dynamic entry/exit framework of Pakes, Ostrovsky, and Berry (POB, 2007), with a fixed-cost formulation that simplifies computation.

3.1 State Variables

Let the market state be

$$s = (n, z),$$

where

- n is the number of incumbent firms at the beginning of the period,
- z is a vector of exogenous profit shifters (market demand and cost conditions).

Exogenous shifters z evolve as a finite-state Markov process. The number of firms n evolves endogenously through entry and exit decisions.

3.2 Short-Run Profits

All active firms in a market share the same *common* profit component:

$$\pi(s; \theta),$$

where θ collects profit function parameters. This is a *reduced-form* profit function that aggregates demand, cost, and the nature of short-run competition.

3.3 Incumbent Problem: Exit vs Continue

Fixed cost draw. At the end of the period, each incumbent draws a private fixed cost λ_i, i.i.d. across firms and time from G_λ.

Value function with discrete choice. Let $VC(s; \theta)$ denote the continuation value (expected value of being an incumbent next period, before drawing next period's fixed cost). The incumbent's value is

$$V(s; \lambda_i, \theta) = \pi(s; \theta) + \max\{\delta VC(s; \theta) - \delta\lambda_i,\ 0\}. \tag{1}$$

Exit rule (threshold). [Exit threshold rule] An incumbent exits if and only if $\lambda_i > VC(s)$.

Proof. From (**??**), the firm compares $\delta VC(s) - \delta\lambda_i$ with 0. Continuing is optimal iff $\delta VC(s) - \delta\lambda_i \geq 0$, i.e. $\lambda_i \leq VC(s)$. Otherwise exit yields the larger payoff. □

Therefore the exit probability is

$$p_x(s; \theta) = \mathbb{P}(\lambda_i > VC(s; \theta)) = 1 - G_\lambda(VC(s; \theta)). \tag{2}$$

3.4 Continuation Value: From the Definition to a Computable Form

Definition. Dropping θ for readability, the continuation value satisfies

$$VC(s) = \mathbb{E}^c_{s'}\Big[\pi(s') + \mathbb{E}_{\lambda'}\big(\max\{\delta VC(s') - \delta\lambda',\, 0\}\big)\Big] \tag{3a}$$
$$= \mathbb{E}^c_{s'}\Big[\pi(s') + \delta(1 - p_x(s'))\big(VC(s') - \mathbb{E}(\lambda' \mid \lambda' \le VC(s'))\big)\Big]. \tag{3b}$$

Here $\mathbb{E}^c_{s'}$ is the expectation over next period states from the perspective of continuing firms.

3.5 Exponential Fixed Costs and the Truncated Mean Formula

The paper assumes $\lambda \sim$ Exponential(mean σ):

$$G_\lambda(\lambda) = 1 - \exp\left(-\frac{\lambda}{\sigma}\right), \quad \lambda \ge 0.$$

[Truncated mean of an exponential] If $\lambda \sim \text{Exp}(\sigma)$, then for any $V > 0$,

$$\mathbb{E}(\lambda \mid \lambda \le V) = \sigma - V \cdot \frac{p_x}{1 - p_x}, \quad \text{where } p_x = \mathbb{P}(\lambda > V) = \exp\left(-\frac{V}{\sigma}\right). \tag{4}$$

Proof. With density $f(\lambda) = \frac{1}{\sigma}e^{-\lambda/\sigma}$ and $1 - p_x = \mathbb{P}(\lambda \le V) = 1 - e^{-V/\sigma}$,

$$\mathbb{E}(\lambda \mid \lambda \le V) = \frac{\int_0^V \lambda \frac{1}{\sigma} e^{-\lambda/\sigma}\, d\lambda}{1 - e^{-V/\sigma}}.$$

Integration by parts gives

$$\int_0^V \lambda \frac{1}{\sigma} e^{-\lambda/\sigma}\, d\lambda = \sigma\big(1 - e^{-V/\sigma}\big) - Ve^{-V/\sigma}.$$

Divide by $1 - e^{-V/\sigma}$ to obtain

$$\mathbb{E}(\lambda \mid \lambda \le V) = \sigma - \frac{Ve^{-V/\sigma}}{1 - e^{-V/\sigma}} = \sigma - V \cdot \frac{p_x}{1 - p_x}.$$

□

Substituting (**??**) into (**??**) yields the simplified continuation value equation:

$$VC(s) = \mathbb{E}^c_{s'}\Big[\pi(s') + \delta VC(s') - \delta\sigma(1 - p_x(s'))\Big]. \tag{5}$$

3.6 Potential Entrant Problem

Each potential entrant draws a private entry cost $\kappa_i \sim G_\kappa$ (i.i.d.). Entry costs are interpreted as startup costs plus costs needed to begin operating next period.

Value of entry.

$$VE(s) = \mathbb{E}^e_{s'}\Big[\pi(s') + \delta VC(s') - \delta\sigma(1 - p_x(s'))\Big], \tag{6}$$

where $\mathbb{E}^e_{s'}$ uses the transition beliefs of entrants.

Entry rule and probability. The entrant enters if $\delta VE(s) \geq \kappa_i$, implying

$$p_e(s) = \mathbb{P}(\kappa_i < \delta VE(s)) = G_\kappa(\delta VE(s)). \tag{7}$$

3.7 Matrix Representation and the POB-Style Solution

Let states be discretized so (n, z) takes finitely many values. Let $\boldsymbol{\pi}, \boldsymbol{VC}, \boldsymbol{p}_x$ be vectors over states. Let M_c be the perceived transition matrix for incumbents.

From (**??**), in vector form:

$$\boldsymbol{VC} = M_c\big[\boldsymbol{\pi} + \delta\boldsymbol{VC} - \delta\sigma(\mathbf{1} - \boldsymbol{p}_x)\big].$$

Rearranging:

$$(\boldsymbol{I} - \delta M_c)\boldsymbol{VC} = M_c\big[\boldsymbol{\pi} - \delta\sigma(\mathbf{1} - \boldsymbol{p}_x)\big],$$

so

$$\boldsymbol{VC} = (\boldsymbol{I} - \delta M_c)^{-1} M_c\big[\boldsymbol{\pi} - \delta\sigma(\mathbf{1} - \boldsymbol{p}_x)\big]. \tag{8}$$

The exit probability satisfies $\mathbf{1} - \boldsymbol{p}_x = G_\lambda(\boldsymbol{VC})$ state-by-state, so $\boldsymbol{VC}$ is solved as a fixed point.

Similarly, with entrant transition matrix M_e,

$$\boldsymbol{VE} = M_e\big[\boldsymbol{\pi} + \delta\boldsymbol{VC} - \delta\sigma(\mathbf{1} - \boldsymbol{p}_x)\big]. \tag{9}$$

4 Empirical Implementation and Estimation Strategy

The estimation proceeds in logically connected steps:

(i) estimate profits $\Rightarrow$ (ii) discretize states and estimate transitions $\Rightarrow$ (iii) back out costs from observed entry/e

4.1 Step 1: Profit Function Estimation

The profit function is:

$$\pi_{mt} = \theta_0 + \sum_{k=1}^{5} \theta_k \mathbf{1}(n_{mt} = k) + \theta_6 n_{mt} + \theta_7 n_{mt}^2 + h(\theta_z, z_{mt}) + f_m + \varepsilon_{mt}. \tag{10}$$

- The dummies $\mathbf{1}(n_{mt} = k)$ flexibly capture discrete differences for $n \leq 5$.
- $\theta_6 n + \theta_7 n^2$ allows the effect of n to continue beyond 5 with diminishing strength.

- $h(\theta_z, z_{mt})$ is a quadratic function of five exogenous shifters:

$$z_{mt} = \{pop_{mt},\ wage_{mt},\ inc_{mt},\ med_{mt},\ mort_{mt}\},$$

  including squares and pairwise interactions.

- f_m is a market fixed effect capturing persistent unobserved heterogeneity.

- The key maintained assumption is that after controlling for (n_{mt}, z_{mt}, f_m), the residual ε_{mt} is i.i.d. and not used by firms to forecast future profits.

4.2 Step 2: State Construction and Transition Matrices

After estimating (**??**), the exogenous shifters are summarized by a scalar index:

$$\widehat{z}_{mt} = h(\widehat{\theta}_z, z_{mt}). \tag{11}$$

Then:

- $\widehat{z}_{mt}$ is discretized into 10 categories.

- estimated fixed effects $\widehat{f}_m$ are discretized into 3 categories.

Thus the empirical state becomes (n, z_d, f_d) and transition matrices M_c and M_e are estimated nonparametrically from observed state-to-state frequencies, exploiting the exogeneity of z and the time-invariance of f.

4.3 Step 3: Estimating Fixed Costs and Entry Costs

Given parameters, one constructs $\widehat{VC}_{mt}(\sigma)$ from (**??**) and $\widehat{VE}_{mt}(\sigma)$ from (**??**). Observed exits and entrants are linked to probabilities through a binomial likelihood:

$$\ell(x_{mt}, e_{mt}; \sigma, \alpha) = (n_{mt} - x_{mt}) \log G_\lambda(\widehat{VC}_{mt}(\sigma); \sigma) + x_{mt} \log\left(1 - G_\lambda(\widehat{VC}_{mt}(\sigma); \sigma)\right) + e_{mt} \log G_\kappa(\widehat{VE}_{mt}(\sigma); \alpha) + (p_{mt} - \tag{12}$$

Summing across markets and time:

$$L(\sigma, \alpha) = \sum_m \sum_t \ell(x_{mt}, e_{mt}; \sigma, \alpha). \tag{13}$$

Distributional choices.

- Fixed costs: $\lambda \sim$ Exponential(mean σ).

- Entry costs: $\kappa \sim \chi^2(\cdot)$ parameterized so that the *single* parameter α equals the mean entry cost.

- For dentists, α is allowed to differ across HPSA vs non-HPSA markets.

TABLE 2 **Number of Potential Entrants (mean across market-time observations)**

Number of Establishments	Dentists: Number of Potential Entrants — Internal Entry Pool	Dentists: Number of Potential Entrants — External Entry Pool	Chiropractors: Number of Potential Entrants — Internal Entry Pool	Chiropractors: Number of Potential Entrants — External Entry Pool
$n = 1$	2.31	23.55	3.42	1.95
$n = 2$	2.74	25.22	3.78	2.88
$n = 3$	3.48	23.41	4.25	4.21
$n = 4$	4.04	23.05	5.13	5.37
$n = 5$	4.75	23.79	5.61	6.83
$n = 6$	6.03	25.45	6.19	7.74
$n = 7$	6.58	27.83	6.16	9.37
$n = 8$	7.81	29.09	8.75	10.67
$n = 9$	8.53	28.26		
$n = 10,11$	9.66	27.13		
$n = 12,13,14$	11.74	25.89		
$n = 15,16,17$	13.83	27.15		
$n = 18,19,20$	15.95	28.21		

Figure 1: TABLE 2 Number of Potential Entrants (mean across market-time observations)

4.4 Potential Entrant Pool p_{mt} and Why It Matters

The probability statement in (12) requires a number of potential entrants p_{mt}. The paper uses two definitions:

- **Internal pool:** maximum number of distinct establishments that ever appear in the market over the sample, minus current incumbents.
- **External pool:** a broader measure that includes non-owner practitioners in nearby counties (capturing spinouts).

This choice matters mainly for the implied *entry rates* and therefore the estimated *entry cost* parameter α.

5 Empirical Results: Tables and Figures (Interpreted as Notes)

This section interprets every table and figure as part of a continuous chain: profits $\Rightarrow$ values $\Rightarrow$ entry/exit probabilities $\Rightarrow$ market structure and counterfactuals.

5.1 Table 2: Number of Potential Entrants

What it reports. Mean potential entrant pools by current number of establishments n.

Dentists.

- Internal pool is small (e.g. 2.31 at $n = 1$; 15.95 at $n = 18$–20).
- External pool is very large (roughly 23–29 even when n is small).

Interpretation: With a larger p_{mt}, the observed e_{mt} implies a smaller entry probability p_e, which (given VE) implies larger estimated entry costs.

TABLE 3 Profit Function Parameter Estimates (standard deviation in parentheses)

Dentist			Chiropractor		
Variable	No Market Fixed Effect	Market Fixed Effect	Variable	No Market Fixed Effect	Market Fixed Effect
Intercept	−11.543 (4.184)*	−2.561 (4.922)	*Intercept*	−1.215 (8.720)	−23.96 (10.55)*
$I(n=1)$	.0379 (.0240)	.0519 (.0301)	$I(n=1)$	.0200 (.0328)	.0613 (.0373)
$I(n=2)$	.0253 (.0173)	.0342 (.0221)	$I(n=2)$	.0211 (.0324)	.0389 (.0373)
$I(n=3)$	.0113 (.0134)	.0179 (.0163)	$I(n=3)$	.0100 (.0328)	.0338 (.0361)
$I(n=4)$	.0112 (.0100)	.0108 (.0122)	$I(n=4)$	.0046 (.0324)	.0192 (.0355)
$I(n=5)$	.0191 (.0087)*	.0154 (.0088)	$I(n=5)$	.0005 (.0331)	.0266 (.0360)
n	−.0044 (.0045)	−.0238 (.0059)*	$I(n=6)$	−.0021 (.0339)	.0041 (.0362)
n^2	.0001 (.0002)	5.55e-4 (2.45e-4)*	$I(n=7)$	−.0277 (.0353)	−.0205 (.0369)
pop	.0127 (.0196)	.0029 (.0301)	pop	−.0097 (.0253)	.0036 (.0403)
pop^2	−6.69e-5 (3.07e-5)*	−1.68e-4 (1.07e-4)	pop^2	−8.92e-5 (2.96e-5)*	−.0001 (.0001)
inc	2.421 (.9027)*	.242 (1.064)	inc	.2004 (1.845)	4.994 (2.248)*
inc^2	−.1260 (.0489)*	.0048 (.0577)	inc^2	−.0062 (.0977)	−.2589 (.1200)*
med	−.0299 (.1005)	.2779 (.1310)*	med	.3042 (.1360)*	.0634 (.2220)
med^2	−.0007 (.0001)*	−.0009 (.0002)*	med^2	−.0004 (.0004)	−.0007 (.0006)
$mort$	.1387 (.0397)*	.1134 (.0363)*	$mort$	−.1040 (.0745)	.0184 (.0801)
$mort^2$	−.0002 (.0001)	−7.97e-5 (1.19e-4)	$mort^2$	.0004 (.0003)	7.62e-5 (2.76e-4)
$wage$	−.1955 (.0577)*	−.0935 (.0554)	$wage$	.1866 (.0687)*	.0867 (.0776)
$wage^2$	−.0013 (.0002)*	−.0008 (.0002)*	$wage^2$	−.0005 (.0001)*	−.0002 (.0001)
$pop \ast w$	2.55e-5 (1.61e-4)	2.67e-4(1.86e-4)	$pop \ast w$	7.91e-6 (9.53e-5)	-2.46e-6 (1.14e-4)
$pop \ast inc$	−.0009 (.0020)	.0019 (.0032)	$pop \ast inc$	.0015 (.0027)	.0005 (.0043)
$pop \ast med$	−.0004 (.0002)*	−.0003 (.0004)	$pop \ast med$	.0004 (.0002)*	.0003 (.0003)
$pop \ast mort$	4.72e-6 (1.18e-3)	5.97e-5 (1.25e-4)	$pop \ast mort$	−.0001 (.0001)	−.0004 (.0001)*
$wage \ast inc$	.0246 (.0062)*	.0119 (.0060)*	$wage \ast inc$	−.0182 (.0072)	−.0090 (.0082)
$wage \ast med$	.0029 (.0006)*	.0023 (.0007)*	$wage \ast med$	.0011 (.0004)*	.0004 (.0005)
$wage \ast mort$	−2.82e-5 (3.09e-4)	.0002 (.0003)	$wage \ast mort$	.0003 (.0004)	.0010 (.0004)*
$inc \ast med$	.0031 (.0107)	−.0267 (.0138)	$inc \ast med$	−.0326 (.0142)*	−.0071 (.0234)
$inc \ast mort$	−.0148 (.0042)*	−.0124 (.0038)*	$inc \ast mort$	.0102 (.0078)	−.0024 (.0084)
$med \ast mort$	−.0003 (.0005)	−.0008 (.0006)	$med \ast mort$	−7.52e-4 (7.80e-4)	.0006 (.0010)
obs	2556	2556	obs	1640	1640
F(27,df)	32.03	58.94	F(27,df)	13.47	5.51
p-value	.0000	.0000	p-value	.0000	.0000

*significant at .05 level

Figure 2: TABLE 3 Profit Function Parameter Estimates (standard deviation in parentheses)

Chiropractors. Internal and external pools are similar in scale, so estimated entry costs are much less sensitive to the pool definition.

5.2 Table 3: Profit Function Parameter Estimates

What it reports. Estimates of (10) with and without market fixed effects.

$$\pi_{mt} = \theta_0 + \sum_{k=1}^{5} \theta_k \mathbf{1}(n_{mt} = k) + \theta_6 n_{mt} + \theta_7 n_{mt}^2 + h(\theta_z, z_{mt}) + f_m + \varepsilon_{mt}. \tag{10}$$

Dentists (key messages).

- Without market fixed effects, the competitive effect of n looks weak (attenuated toward zero).
- With market fixed effects, the competitive effect becomes strong: $\theta_6 < 0$ and $\theta_7 > 0$ are statistically significant, implying profits decline with n but at a diminishing rate for higher n.
- The paper summarizes these estimates as large *profit premia* in small-n markets: relative to a 5-firm market, monopoly/duopoly/triopoly/4-firm premia are approximately 69%, 47%, 26%, and 10%.

TABLE 4 Fixed Cost and Entry Cost Parameter Estimates (standard errors in parentheses)

	Maximum Likelihood Estimator			GMM Estimator		
Panel A. Dentist (all markets)						
Entry Pool	σ	α		σ	α	
Internal	0.373 (0.006)	2.003 (0.013)		0.362 (0.004)	2.073 (0.031)	
External	0.375 (0.006)	3.299 (0.039)		0.362 (0.004)	2.644 (0.067)	
Panel B. Dentist (HPSA versus non-HPSA markets)						
Entry Pool	σ	α (HPSA)	α (non-HPSA)	σ	α (HPSA)	α (non-HPSA)
Internal	0.366 (0.009)	1.797 (0.069)	2.019 (0.041)	0.351 (0.005)	1.877 (0.076)	2.098 (0.032)
External	0.368 (0.008)	3.083 (0.169)	3.376 (0.079)	0.351 (0.005)	1.943 (0.213)	2.695 (0.092)
Panel C. Chiropractor						
Entry Pool	σ	α		σ	α	
Internal	0.275 (0.005)	1.367 (0.015)		0.254 (0.004)	1.337 (0.023)	
External	0.274 (0.005)	1.302 (0.022)		0.254 (0.004)	1.302 (0.028)	

Figure 3: TABLE 4 Fixed Cost and Entry Cost Parameters

Chiropractors (key messages).

- Profit declines with n are much smaller and mostly not statistically significant in the fixed-effect specification.
- Reported premia relative to 5-firm markets are modest (around 15%, 5%, 3%, 3% for $n = 1, 2, 3, 4$).

Interpretation for identification. This table provides the model's measure of **toughness of competition**: the slope of profits with respect to the number of firms, after controlling for persistent market heterogeneity.

5.3 Table 4: Fixed Cost and Entry Cost Parameters

What it reports. Estimated means of cost distributions (in millions of 1983 dollars, over five-year intervals), using ML and GMM, and using internal vs external entry pools.

Fixed costs σ.

- Dentists: $\sigma \approx 0.37$ (stable across specifications).
- Chiropractors: $\sigma \approx 0.275$ (lower than dentists).

Interpretation: Dentists face higher operating/fixed costs on average, which ceteris paribus increases exit incentives, but this must be compared to their typically higher profits/values.

Entry costs α and HPSA subsidy evidence.

- Dentists (internal pool): $\alpha_{\text{HPSA}} \approx 1.797$ vs $\alpha_{\text{non-HPSA}} \approx 2.019$.
- This is an **11% lower mean entry cost** in HPSA markets, consistent with entry subsidies.
- Dentists (external pool): α rises substantially because the pool is much larger, implying lower inferred entry probabilities.

TABLE 5 **Predicted Value of Dynamic Benefits VC, VE (evaluated at different values of the state variables) (millions of 1983 dollars)**

	VC for Incumbents — Dentist			*VE* for Potential Entrants — Dentist		
	Low(z, f)	Mid(z, f)	High(z, f)	Low(z, f)	Mid(z, f)	High(z, f)
$n = 1$	0.433	0.764	1.286	0.394	0.722	1.247
$n = 2$	.0383	0.714	1.236	0.350	0.678	1.202
$n = 3$	0.331	0.661	1.184	0.304	0.632	1.156
$n = 4$	0.297	0.628	1.150	0.273	0.601	1.126
$n = 5$	0.261	0.591	1.114	0.240	0.568	1.093
$n = 6$	0.236	0.567	1.089	0.218	0.546	1.070
$n = 8$	0.195	0.525	1.048	0.180	0.508	1.032
$n = 12$	0.126	0.457	0.979	0.117	0.445	0.969
$n = 16$	0.089	0.420	0.942	0.083	0.412	0.936
$n = 20$	0.067	0.397	0.920	0.064	0.392	0.916
	VC for Incumbents — Chiro			*VE* for Potential Entrants — Chiro		
$n = 1$	0.178	0.344	0.562	0.170	0.335	0.553
$n = 2$	0.166	0.332	0.551	0.161	0.326	0.544
$n = 3$	0.155	0.321	0.540	0.151	0.316	0.534
$n = 4$	0.148	0.314	0.532	0.144	0.308	0.527
$n = 5$	0.138	0.304	0.522	0.134	0.299	0.517
$n = 6$	0.132	0.298	0.516	0.129	0.294	0.512
$n = 7$	0.127	0.294	0.512	0.126	0.291	0.509
$n = 8$	0.123	0.289	0.508	0.123	0.287	0.506

Figure 4: TABLE5 Predicted Continuation Value V C and Entry Value V E

- Chiropractors: $\alpha \approx 1.3$–1.37 (lower than dentists), with little sensitivity to pool definition.

5.4 Table 5: Predicted Continuation Value VC and Entry Value VE

What it reports. $VC(n, z, f)$ and $VE(n, z, f)$ evaluated at low/mid/high values of (z, f).

Dentists.

- Values are high and sensitive to exogenous conditions: at $n = 1$,

$$VC \in \{0.433,\ 0.764,\ 1.286\}.$$

- Holding (z, f) fixed, VC declines with n (competition plus endogenous dynamics). For low (z, f), VC falls from 0.433 ($n = 1$) to 0.067 ($n = 20$).
- VE is close to VC in each state, reflecting similar entrant vs incumbent transition beliefs in the application.

Chiropractors.

- Values are substantially lower: at $n = 1$ and high (z, f), $VC = 0.562$ (vs 1.286 for dentists).
- The decline with n is mild (e.g. high (z, f): VC from 0.562 at $n = 1$ to 0.508 at $n = 8$), consistent with weaker competitive effects in Table 3.

VALUE OF CONTINUATION $VC(n, z, f)$ LOW, MEDIUM, AND HIGH VALUES OF (z, f)

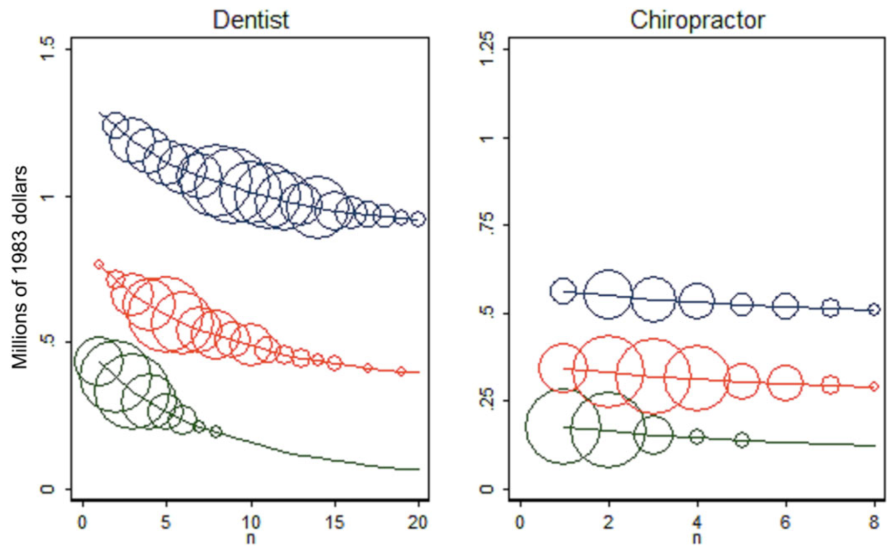

Figure 5: FIGURE 1 Value of Continuation Across States

Interpretation. Table 5 translates profit and cost primitives into **long-run firm values**. It also shows that cross-market heterogeneity in (z, f) can matter more than incremental changes in n.

5.5 Figure 1: Value of Continuation Across States

What it shows. A graphical representation of $VC(n)$ for low/mid/high (z, f), with circle sizes indicating how often each state appears in the data.

Interpretation.

- The vertical separation between low/mid/high curves emphasizes the importance of exogenous heterogeneity for long-run firm values.
- Within each curve, VC declines with n, capturing endogenous competitive pressure.
- The distribution of circles shows that high-profit states tend to support more firms, but there is still wide dispersion in n within a given (z, f) region, motivating the role of entry/exit dynamics and unobserved heterogeneity.

5.6 Table 6: Predicted Probabilities of Exit and Entry

What it reports. $p_x(n, z, f)$ and $p_e(n, z, f)$ implied by the estimated model.

Dentists.

- Exit probability increases with n and decreases with (z, f).

TABLE 6 **Predicted Probabilities of Exit and Entry (evaluated at different values of the state variables)**

	Probability of Exit — Dentist			Probability of Entry — Dentist		
	Low(z, f)	Mid(z, f)	High(z, f)	Low(z, f)	Mid(z, f)	High(z, f)
$n = 1$	0.313	0.129	0.032	0.141	0.216	0.382
$n = 2$	0.358	0.148	0.036	0.126	0.204	0.371
$n = 3$	0.412	0.170	0.042	0.110	0.191	0.360
$n = 4$	0.451	0.186	0.046	0.100	0.182	0.352
$n = 5$	0.497	0.205	0.050	0.088	0.173	0.344
$n = 6$	0.531	0.219	0.054	0.080	0.166	0.338
$n = 8$	0.593	0.244	0.060	0.067	0.155	0.328
$n = 12$	0.713	0.294	0.072	0.044	0.136	0.312
$n = 16$	0.787	0.324	0.080	0.032	0.124	0.303
$n = 20$	0.836	0.345	0.085	0.024	0.117	0.297
	Probability of Exit — Chiro			Probability of Entry — Chiro		
$n = 1$	0.524	0.286	0.129	0.133	0.245	0.371
$n = 2$	0.547	0.299	0.135	0.127	0.239	0.367
$n = 3$	0.569	0.311	0.141	0.119	0.233	0.362
$n = 4$	0.585	0.319	0.144	0.114	0.228	0.358
$n = 5$	0.606	0.331	0.150	0.107	0.222	0.352
$n = 6$	0.620	0.339	0.153	0.103	0.219	0.350
$n = 7$	0.629	0.344	0.155	0.101	0.217	0.348
$n = 8$	0.639	0.349	0.158	0.098	0.215	0.346

Figure 6: TABLE 6 Predicted Probabilities of Exit and Entry

- It ranges from very low (e.g. $p_x = 0.032$ at $n = 1$ and high (z, f)) to extremely high (e.g. $p_x = 0.836$ at $n = 20$ and low (z, f)).
- Entry probability decreases with n and increases with (z, f): from $p_e = 0.382$ (monopoly, high (z, f)) down to $p_e = 0.024$ (20 firms, low (z, f)).

Chiropractors.

- Exit probabilities are generally higher than dentists (e.g. $p_x = 0.524$ at $n = 1$ and low (z, f)), consistent with lower values in Table 5.
- Entry probabilities are of similar magnitude to dentists in small-n states (e.g. $p_e = 0.371$ at $n = 1$ and high (z, f)), helping explain why chiropractors show high turnover in Table 1.

Logical link. Table 6 is the model's mechanism connecting firm values (Table 5) to observed turnover (Table 1).

5.7 Tables 7 and 8: Model Fit for Dental Market Structure

These tables test whether the model reproduces market structure patterns when simulated forward.

Table 7 (distribution of n).

- The simulated model matches the empirical distribution of the number of firms in both non-HPSA and HPSA markets.
- It reproduces the fact that HPSA markets tend to have fewer firms.

TABLE 7 **Distribution of the Number of Dental Establishments**

	non-HPSA Markets		HPSA Markets	
Number of Establishments	Data	Model	Data	Model
$n = 1$	.018	.043	.034	.059
$n = (2,3)$	.166	.162	.314	.268
$n = (4,5)$	.223	.209	.275	.251
$n = (6,7,8,9,10)$	.376	.382	.305	.340
$n > 10$	.217	.204	.072	.081

TABLE 8 **Average Number of Dental Establishments Per Market**

	non-HPSA Markets		HPSA Markets	
z Category	Data	Model	Data	Model
1	3.83	3.80	4.13	4.35
2	4.75	4.36	4.29	4.31
3	4.89	5.03	4.71	4.36
4	5.85	5.66	4.79	4.27
5	6.07	5.96	5.25	5.05
6	7.03	6.85	4.58	5.11
7	7.89	7.40	5.63	5.71
8	8.93	8.24	8.71	7.28
9	10.27	9.52	9.17	8.61
10	13.18	11.72	13.09	11.94

Figure 7: TABLE7 and TABLE8

Table 8 (mean n by z category).

- Average n increases with the exogenous profit index category z, both in data and in the model.
- The model slightly underpredicts the number of firms for higher z categories in non-HPSA markets, but the monotone pattern is captured.

Interpretation. Together, Tables 7–8 provide evidence that the estimated primitives are sufficient for the model to replicate key cross-market structure facts.

5.8 Tables 9 and 10: Counterfactual—Reducing Entry Costs (HPSA-Style Subsidy)

The counterfactual shifts mean entry costs from non-HPSA levels to HPSA levels:

$$\alpha : 2.019 \to 1.797 \quad \text{(about 11\% reduction).}$$

Table 9 (impact on entrants).

- VE falls slightly (more expected competition in the future), with larger percentage declines in low-profit states.
- Entry probabilities increase sharply: roughly 12%–31% increases depending on the state, with the largest percentage gains in low (z, f) markets.

TABLE 9 Reduction in Entry Cost: Impact on Entrants (percentage change in the variable)

Number of Firms	$VE(n,z,f)$ Low (z,f)	$VE(n,z,f)$ Mid (z,f)	$VE(n,z,f)$ High (z,f)	$p^e(n,z,f)$ Low (z,f)	$p^e(n,z,f)$ Mid (z,f)	$p^e(n,z,f)$ High (z,f)
$n=1$	−5.83	−3.70	−2.10	20.30	15.88	11.87
$n=2$	−5.60	−3.44	−1.89	21.90	16.79	12.38
$n=3$	−5.97	−3.47	−1.84	23.12	17.51	12.79
$n=4$	−5.84	−3.28	−1.70	24.53	18.23	13.17
$n=5$	−6.09	−3.23	−1.63	25.80	18.89	13.52
$n=7$	−5.86	−2.92	−1.41	28.44	20.06	14.10
$n=9$	−5.62	−2.63	−1.22	31.15	21.11	14.59

TABLE 10 Reduction in Entry Cost: Impact on Incumbent Establishments (percentage change in the variable)

Number of Firms	$VC(n,z,f)$ Low (z,f)	$VC(n,z,f)$ Mid (z,f)	$VC(n,z,f)$ High (z,f)	$p^x(n,z,f)$ Low (z,f)	$p^x(n,z,f)$ Mid (z,f)	$p^x(n,z,f)$ High (z,f)
$n=1$	−6.50	−4.26	−2.50	7.85	9.11	8.99
$n=2$	−6.26	−3.97	−2.26	6.64	7.89	7.76
$n=3$	−6.50	−3.91	−2.15	5.93	7.18	7.05
$n=4$	−6.36	−3.71	−1.98	5.20	6.44	6.31
$n=5$	−6.62	−3.66	−1.90	4.73	5.97	5.84
$n=7$	−6.31	−3.28	−1.63	3.69	4.91	4.78
$n=9$	−6.06	−2.97	−1.42	2.92	4.13	4.01

Figure 8: TABLE 9 and TABLE 10

Table 10 (impact on incumbents).

- VC falls (incumbents expect more entry and tougher competition).
- Exit probabilities increase (typically a few percentage points up to nearly 9% in some states).

Core economic lesson from Tables 9–10. An entry subsidy can increase entry, but it can also **raise exit** and **lower incumbent values**. Therefore, the net change in the number of firms is not the entry change alone.

5.9 Table 11: Cost–Benefit Comparison of Alternative Policies

Table 11 compares:

- **Benchmark** (non-HPSA costs),
- **Entry cost reduction** (HPSA-style),
- **Fixed cost reduction** (operating subsidy),
- **Expanded entry subsidy program**.

Market structure changes. Both the entry subsidy and a modest fixed-cost subsidy reduce the fraction of markets with few firms (e.g. $Pr(n \leq 5)$ falls from 0.592 to 0.562 or 0.571).

Different adjustment mechanisms.

- Entry cost reduction: raises entrants a lot (from 1.396 to 1.657) but also raises exits (from 1.029 to 1.131), netting a moderate increase in firms (+0.526 per market).

TABLE 11 Cost-Benefit Comparison of Alternative Policies

Impact on Market Structure	Benchmark non-HPSA costs	Entry Cost Reduction	Fixed Cost Reduction	Expand Program
Pr($n = 1$)	0.062	0.055	0.056	0.034
Pr($n \leq 3$)	0.338	0.313	0.319	0.246
Pr($n \leq 5$)	0.592	0.562	0.571	0.475
Average number of entrants/market	1.396	1.657	1.423	2.563
Average number of exits/market	1.029	1.131	0.950	1.477
Net change in establishments/market	0.367	0.526	0.473	1.086
Cost/additional entrant (millions 1983 $)		0.103		0.075
Cost/additional establishment (millions 1983 $)		0.170	0.503	0.140

Figure 9: TABLE11

- Fixed cost reduction: reduces exit substantially, has little effect on entry, netting a similar increase in firms (+0.473 per market).

Cost effectiveness.

- Entry subsidy: cost per additional establishment $\approx$ 0.170 million (1983 dollars).
- Fixed-cost subsidy: cost per additional establishment $\approx$ 0.503 million, higher because it pays incumbents who would have stayed anyway.

Program expansion. A larger entry subsidy (mean entry cost down to 1.3) produces a much larger increase in market structure (net change +1.086 firms per market) with both higher entry and higher exit.

6 Appendix Model: Infinite Potential Entrants and Poisson Entry

The baseline model requires specifying p_{mt} and an entry cost distribution G_κ. The appendix provides an alternative approach: treat the pool of potential entrants as very large, and model entry as a Poisson process with a free-entry condition.

6.1 Free-Entry Condition

Let N^e be the number of potential entrants, each enters with probability p_N. Given state (n, z), if $\tilde{e}$ other potential entrants enter, the value of entry is

$$VE(n, z; \tilde{e}) = \mathbb{E}^e_{x,z'}\Big[\pi(n + \tilde{e} + 1 - x, z') + \mathbb{E}_{\lambda'} \max\{\delta VC(n + \tilde{e} + 1 - x, z') - \delta\lambda', 0\}\Big].$$

A symmetric mixed-strategy equilibrium satisfies:

$$\sum_{\tilde{e}=0}^{N^e-1} \binom{N^e - 1}{\tilde{e}} (p_N)^{\tilde{e}} (1 - p_N)^{N^e-1-\tilde{e}} VE(n, z; \tilde{e}) = \kappa(n, z). \tag{A1}$$

As $N^e \to \infty$ with small p_N, the number of entrants becomes approximately Poisson with mean $\gamma(n, z)$.

6.2 Poisson Formulation for the Value of Entry

With $e \sim \text{Poisson}(\gamma(n, z))$, the value of entry becomes

$$VE(n,z) = \sum_{e=1}^{\infty} \frac{\gamma(n,z)^e}{e!} e^{-\gamma(n,z)} \Big(\mathbb{E}^e_{x,z'} \big[\pi(n+e-x, z') + \mathbb{E}_{\lambda'} \max\{\delta VC(n+e-x, z') - \delta\lambda', 0\} \big] \Big) = \kappa(n,z). \tag{A2}$$

Thus $VE(n, z)$ is directly interpretable as the entry cost implied by free entry.

6.3 Estimating the Poisson Mean $\gamma(n, z)$

The paper uses a Poisson regression with market fixed effects:

$$\mathbb{E}(e_{mt} \mid \gamma(n_{mt}, z_{mt}), \gamma_m) = \gamma_m \gamma(n_{mt}, z_{mt}) = \gamma_m \exp\Big(\gamma_0 + \sum_{k=1}^{5} \gamma_k \mathbf{1}(n_{mt} = k) + \gamma_6 n_{mt} + \gamma_7 n_{mt}^2 + \gamma_8 z_{mt}\Big). \tag{A3}$$

6.4 Simulating the Value of Entry

Using simulated entrant counts e^s_{mt}:

$$VE^S(n_{mt}, z_{mt}) = \frac{1}{S} \sum_{s=1}^{S} \mathbb{E}_{x,z'} \Big[\pi(n_{mt} + e^s_{mt} - x, z') + \mathbb{E}_{\lambda'} \max\{\delta \widehat{VC}(n_{mt} + e^s_{mt} - x, z') - \delta\lambda', 0\} \Big]. \tag{A4}$$

By free entry, this is interpreted as $\kappa(n, z)$.

6.5 Table A1: Poisson Regression Estimates

What it reports. Coefficients of (**??**).

- Indicators for small n are positive and decline with n: entry is highest when the market currently has few incumbents.
- The coefficient on n is negative: holding other things fixed, more incumbents discourages entry.
- The coefficient on z is positive: more favorable exogenous profitability conditions increase entry.

6.6 Figure A1: Kernel Density of $VE(n, z)$ Across Markets

What it shows. The distribution of $VE(n, z)$ implied by:

- the benchmark finite-entrant-pool model, and
- the Poisson/free-entry model.

Interpretation: The two densities are very similar, suggesting that in this application, assumptions about the size of the potential entrant pool do not materially change the implied long-run payoff to entry (even though they can matter for the inferred entry-cost distribution parameter in the benchmark model).

7 Clean Takeaways

1. **Three primitives determine market structure dynamically:** (i) profit function slope in n (competition), (ii) fixed costs (exit), (iii) entry costs (entry).
2. **Dynamic values matter:** firms choose based on VC and VE, not just current profits.
3. **Entry subsidies are not mechanically net entry:** they can increase exit and lower incumbent values.
4. **Operating subsidies work through exit:** they reduce exit but are expensive because many inframarginal incumbents receive payments.
5. **Entry pool assumptions:** they can strongly affect inferred entry costs, motivating alternative Poisson/free-entry modeling.